METRIC

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SECTION 073 - PART 1
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MILITARY STANDARD

INTERFACE STANDARD FOR SHIPBOARD SYSTEMS

SECTION 073 - PART 1

VIBRATION ENVIRONMENT

(METRIC)

FSC 1990

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DEPARTMENT OF THE NAVY

WASHINGTON, DC 20362

Interface Standard for Shipboard Systems,
Vibration Environment

DOD-STD-1399(NAVY)
SECTION 073 - Part 1

1. This Military Standard is approved for use by all interested Commands of the Department of the Navy in the technical development plans, design, and procurement specifications for new ship acquisitions, ship modernizations or conversions, and equipment for installation therein and into active fleet ships where applicable, and is available for use by all Departments and Agencies of the Department of Defense.

2. Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Ship Engineering Center, SEC 6124, Department of the Navy, Washington, DC 20362, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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FOREWORD

Purpose. This section defines the interface requirements for shipboard equipment, and the constraints imposed on design, location, and installation by the shipboard vibration environment, to ensure that such equipment will be compatible with that environment.

Nature of the interface. Ships are subject to vibration generated by mechanical, hydrodynamical, and operational forces or factors. The resultant vibration environment has variable characteristics with potentially degrading effects on performance of shipboard equipment and on efficiency of personnel. To reduce the potentially adverse effects of the vibration environment, test standards for equipment have been established; also, the effects of vibration can be minimized by design, layout, and installation considerations.

Structure. The technical content identifies qualitatively the various elements that contribute to vibration generation and affect vibration responses on board ship. Constraints on vibration testing requirements and on design, location, and installation factors are then established.

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1. GENERAL, SCOPE, INTERFACE, AND APPLICABILITY

1.1 General. Policies and procedures established by DOD-STD-1399 are mandatory. This section and the basic standard are to be viewed as an integral single document.

1.2 Scope. This section establishes requirements for equipment whose performance could be adversely affected by shipboard vibration to ensure compatibility between such equipment and the vibration environment interface characteristics.

1.3 Interface. Basic characteristic and constraint categories concerned with this interface are shown symbolically on figure 1 (see section 3 "Definitions" of DOD-STD-1399):

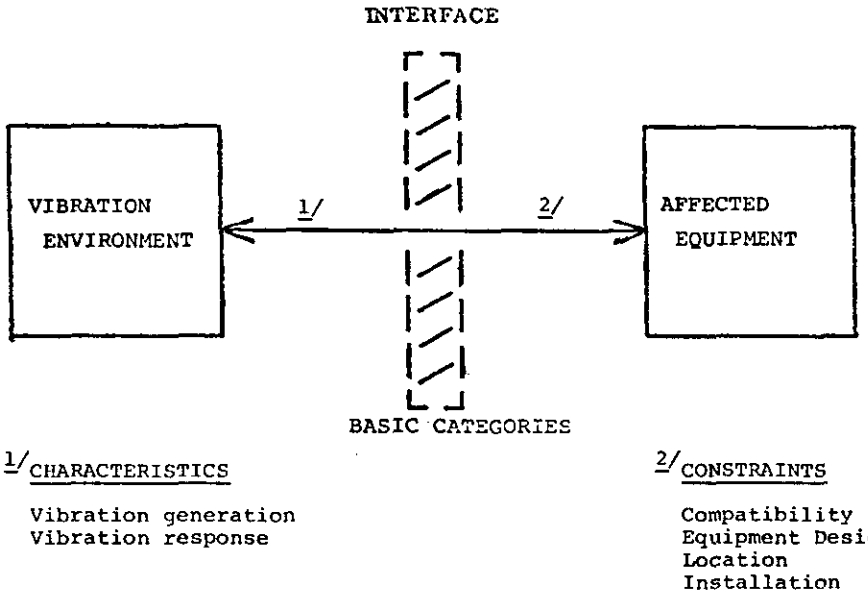


FIGURE 1. Interface.

The particular interface characteristics and constraints pertinent to this section are described in 5.2 and 5.3.

1.4 Applicability. This section applies to the external vibration environment imposed on shipboard equipment. It does not apply to internally excited, torsional, longitudinal, or lateral vibrations generated by individual machinery systems, except to the extent they might contribute to the overall vibration environment (see 3.3). The criteria of this section shall be used in the design or selection of all shipboard equipment whose performance may be affected by this environment (see 5.3.1).

2. REFERENCED DOCUMENTS

2.1 Issues of documents. The following document of the issue in effect on date of invitation for bids or request for proposal, forms a part of this standard to the extent specified herein.

STANDARD

MILITARY

MIL-STD-167-1 - Mechanical Vibrations of Shipboard Equipment (Type I - Environmental and Type II - Internally Excited).

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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3. DEFINITIONS

3.1 Equipment. Equipment, as used in this section, is a physical unit which is installed and operated on board ship and which can be tested on a machine conforming to the type I test conditions specified in MIL-STD-167-1.

3.2 Maximum exciting frequency. Maximum exciting frequency is defined as the ship's full power blade rate. The full power blade rate is the design full power shaft revolutions per minute (r/min) times the number of propeller blades divided by 60 and expressed in Hertz (Hz).

3.3 Vibration environment. Vibration environment is a response of the ship hull, masts, structures, and appendages to the combined effect of those mechanical, hydrodynamical, and operational forces which act upon or within the ship in such manner as to produce reactive vibration forces.

4. REQUIREMENTS

4.1 Requirements. The specific interface requirements and constraints established herein are mandatory and shall be adhered to by SYSCOMs, Project managers, contractors, and all others engaged in any aspect of total shipboard design to which these requirements and constraints apply, including equipment design, production, and installation (see section 4 "Requirements" of DOD-STD-1399).

5. INTERFACE CHARACTERISTICS AND CONSTRAINTS

5.1 General considerations. Vibration is experienced by all ships in varying degree and the effects on equipment installed therein are dependent upon many complex factors such as individual design, location in the ship, method and type of mounting. This vibration, experienced both in a long-term cumulative mode and as intermittent influences, can have a degrading effect on equipment reliability ranging from reduced performance to catastrophic failure. Sustained vibration may also have an adverse effect on the efficiency of ship personnel.

5.1.1 Response characteristics. Ship response characteristics which affect the vibration environment are forced into reaction by complex ship dynamics which include mechanical, hydrodynamical, and operational driving forces. Many kinds of vibration patterns and mode couplings are produced, and these will differ at various ship locations due to structural design and configurations. The combined effects of shaft and machinery imbalance, weapon firing, variable thrust and reaction torque at blade frequency, and ship motion in the ocean environment contribute to such diverse vibration patterns. Therefore, equipment mounted on the hull or superstructure may be exposed to different and varying vibration environments.

5.1.2 Expected vibration levels. The vibration levels which logically may be expected during the life of a ship do not lend themselves to direct measurement or reasonably accurate prediction, nor can the expected environment be precisely simulated in a practical manner to validate equipment reaction. Therefore, it is necessary to postulate empirical parameters of the vibration environment, and to establish practical test methods and procedures which, while they do not provide directly the precise effects of long term vibration on equipment, will furnish guidance for the design of compatible equipment. The primary goal is to provide assurance that the ship can perform its assigned mission during its expected life cycle when prudently operated under adverse sea conditions. Vibration levels cannot be completely suppressed, but can be controlled to tolerable limits during the ship design phase..

5.2 Interface characteristics. Interface characteristics of the ship's vibration environment are described in 5.2.1. In varying degrees, these characteristics will be present on all ships, and will have an effect on equipment installed therein. Such characteristics are described qualitatively since quantitative values for particular shipboard situations are not readily obtainable and the ultimate installation location of equipment may not be known. Therefore, general empirical criteria of the vibration environment related to "worst case" conditions which may reasonably be expected have been established (see 5.3.1).

5.2.1 Elements of vibration environment. The elements of vibration environment comprise a large body of individual items, each contributing in its own way to the resulting complex environment. A broad band of frequencies is involved, but for practical purposes the range considered for Navy shipboard applications is limited to 4 Hz to 50 Hz. Table I represents a qualitative summary of the contributing factors and the vibrational responses involved.

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TABLE I. Shipboard vibration elements.

Vibration generation	Vibration responses
Mechanical	*Types of vibration*
Rotating machinery	Longitudinal
Fluctuating torque	Flexural
Unbalance	Torsional
Reciprocating machinery	Whirl
Shaft bearing forces	Coupled modes
Electromagnetic systems	
Control systems	Structural configurations
Hydrodynamical	Hull girder
	Bulkheads and decks
Propeller forces	Superstructure
Seaway	Mast
Splashes and impulses	Protuberances
Structural recesses	Appendages
Hydroelastic effects	Propellers
Vortices	Machinery systems
Fluid flow in pipes	Weapons systems
	Electronic systems
Operational	Transmitted by
Maneuvers	
Shallow water	Foundations
Weapon launching	Mounts
Weapon training	Tuners
Shocks	Shafts
Blasts	Beams
Crew activity	Couplings
	Dampers
	Pipes and hangers

5.3 Interface constraints. The interface characteristics of the shipboard vibration environment impose certain constraints on the design, location, and installation of equipment to ensure compatibility between such equipment and the vibration environment. These constraints are described in 5.3.1 through 5.3.4.

5.3.1 Equipment design. All equipment destined for shipboard use shall be designed or selected and then tested for conformance to the type I requirements of MIL-STD-167-1. This is the "worst case" environment (see 5.2).

5.3.1.1 Known ship installations. When equipment is to be installed on a single class of ships or on a particular ship, the required equipment vibration endurance test, as prescribed in table I of MIL-STD-167-1, need only be performed up through the frequency range which includes the maximum exciting frequency.

5.3.2 Subsystem installations. Equipment-associated support fittings such as piping, piping hangers, waveguides, and cabling become an integral part of the emplaced equipment and are mutually affected by vibration responses. Such subsystems shall be designed with a natural frequency no less than the maximum exciting frequency.

5.3.3 Location criteria. Shipboard units shall not be exempted from the requirements of this standard on the basis of planned location in an area of the ship which is predictably free of vibration.

5.3.4 Installation considerations. The following installation considerations shall be applied:

(a) Vibration isolation mounts or distributed isolation material (DIM) shall not be used except for specific applications approved by the command or agency concerned.

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(b) When resilient mounts are allowed, sufficient clearance shall be provided to prevent equipment from striking the adjacent structure or equipment during maximum excursions. Flexible connections and suitable electrical grounding shall also be provided.

(c) If two or more equipment units are required to be mounted on resilient mountings and are connected by shafting, such units shall be mounted on a common subbase with the resilient mountings installed between the subbase and the ship structure.

6. DOCUMENTATION REQUIREMENTS

6.1 DD Form 1423. When this standard is invoked, a test report conforming to type I requirements of MIL-STD-167-1 shall be specified by the Principal Development Activity (PDA) on DD Form 1423 (Contract Data Requirements List) attached to the contract or order. The Naval Ship Engineering Center, Ship Noise and Vibration Control Section, shall be included in the distribution listed in Block 14 of DD Form 1423 for data specified.

7. DEVIATIONS

7.1 Conditions. In achieving the purpose of this section it is recognized that there must be some flexibility of application. During the early design stage of shipboard equipment which will be exposed to the shipboard vibration environment it may become apparent that significant advantages in the overall design/operation of such equipment can be achieved by deviating from the standard characteristics specified herein. In such instance, the provisions of section 6 "Deviations" of DOD-STD-1399 must be complied with.

7.1.1 Deviation procedure. Requests for deviations shall be submitted to the Naval Sea Systems Command with copies to:

(a) Program/Project Manager.
(b) NAVSEC 6105.

Review activities:
EC, OS, AS
User activities:
YD, CG

Preparing activity:
Navy - SH
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